AIAS+ 2026 · ORAL PRESENTATION

# FlyGM

## Whole-Brain Connectomic Graph Model for Whole-Body Locomotion Control

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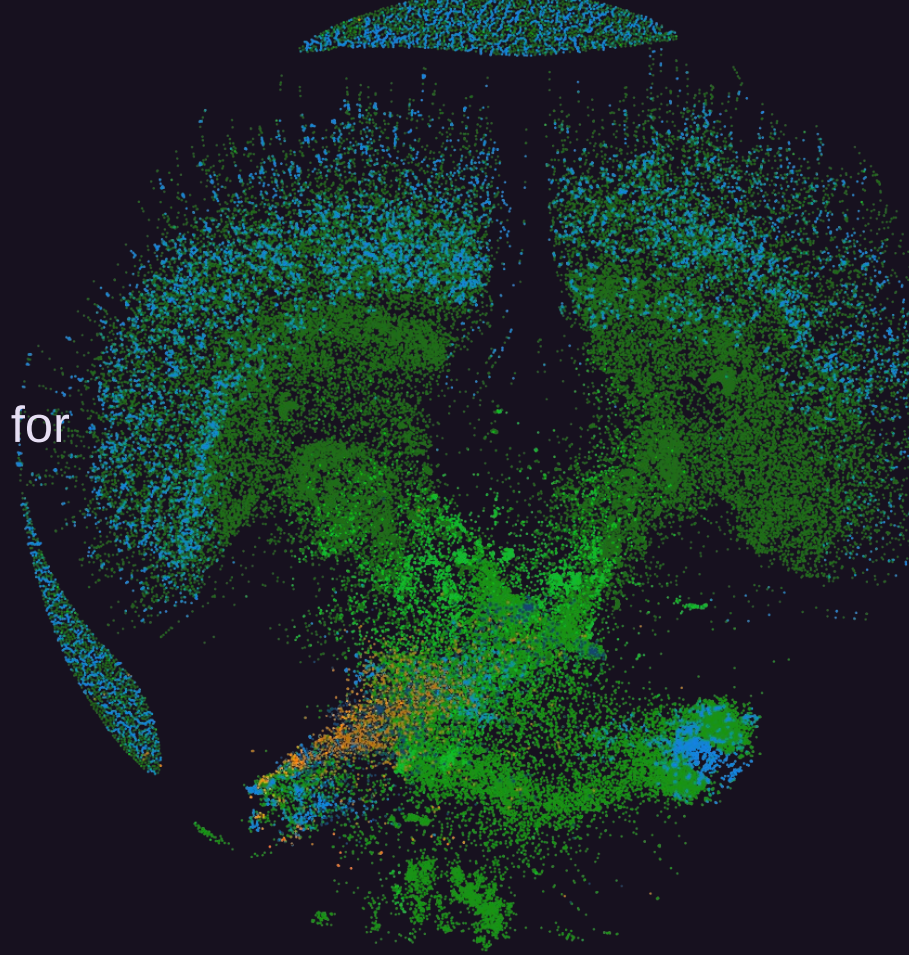
Tsinghua University · LNS Group

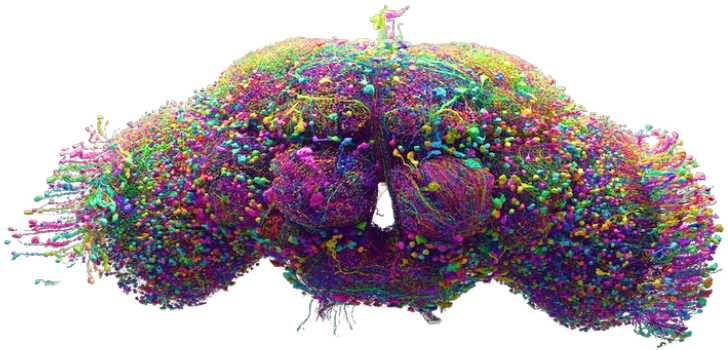


LNS Group

Demo page

WHY NOW

# The fly brain and the fly body are now both digital



**139,255** neurons

**54.5 M** synapses

**8,453** cell types

**2024**

## FlyWire: a complete adult brain

Synapse-resolution wiring diagram of one female fly, proofread by a global community [2, 3]

**2024**

## Connectome-constrained models

Predict visual-system activity [5] and whole-brain sensorimotor responses [4]

**2025**

## flybody: a whole-body fly in physics

Anatomically detailed MuJoCo body; learned controllers walk and fly [7]; see also NeuroMechFly v2 [6]

**2026**

## Male CNS: brain + nerve cord

166,700 neurons, brain and ventral nerve cord joined at the neck [8]

Brain rendering: FlyWire [2]. Statistics: FlyWire v783 [2, 3]; male CNS v1.0 release [8].

THE 2026 DEBATE

# Can a wiring diagram, by itself, drive a body?

## WHAT WENT VIRAL

### “We’ve uploaded a fruit fly.”

**Eon Systems, March 2026 [9]**

A connectome-based LIF brain model [4] coupled to a NeuroMechFly body [6]: it walks, grooms and feeds.

### “No, we haven’t uploaded a fly yet.”

**A. Zeleznikow-Johnston, LessWrong, March 2026 [10]**

Asks for evidence beyond a video: data, a write-up, known neural signatures.

### “Deconstructing viral fly sims”

**P. Mineault, NeuroAI, 2026 [11]**

Most of the behavior comes from the body model's built-in controllers, not the connectome.

## WHAT THE WIRING DIAGRAM LEAVES OUT

### Synaptic strength and sign in vivo

The connectome gives possible paths, not how strongly neurons actually influence each other [12].

### Neuron dynamics

Emulations assume simplified, largely uniform neuron models for very different cell types [4].

### The interface to the body

In the demo, a few hand-picked neurons drive the body model's built-in locomotion and feeding controllers [9, 11].

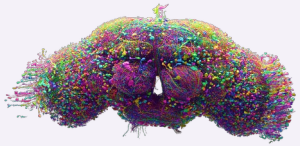
OPEN QUESTION

**How should a connectome control a whole body?**

Public discussion summarized from the sources cited; quotes are headlines of [9–11].

OUR APPROACH

# FlyGM: use the wiring as the controller's architecture



## Emulate the wiring

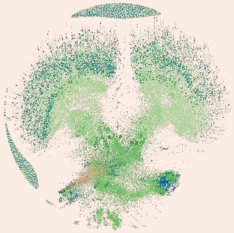
+ Biological connectome, no training
− Generic neuron models
− Body control handed to built-in controllers

*e.g. [4, 9]*

## Learn a generic network

+ Realistic whole-body walking and flight
− Architecture unrelated to the brain
− Hard to relate to biological circuits

*e.g. flybody [7]*



## FlyGM (ours)

+ Motor commands pass through the 139k-neuron connectome
+ Trained end-to-end on whole-body tasks
→ Tests whether real wiring helps

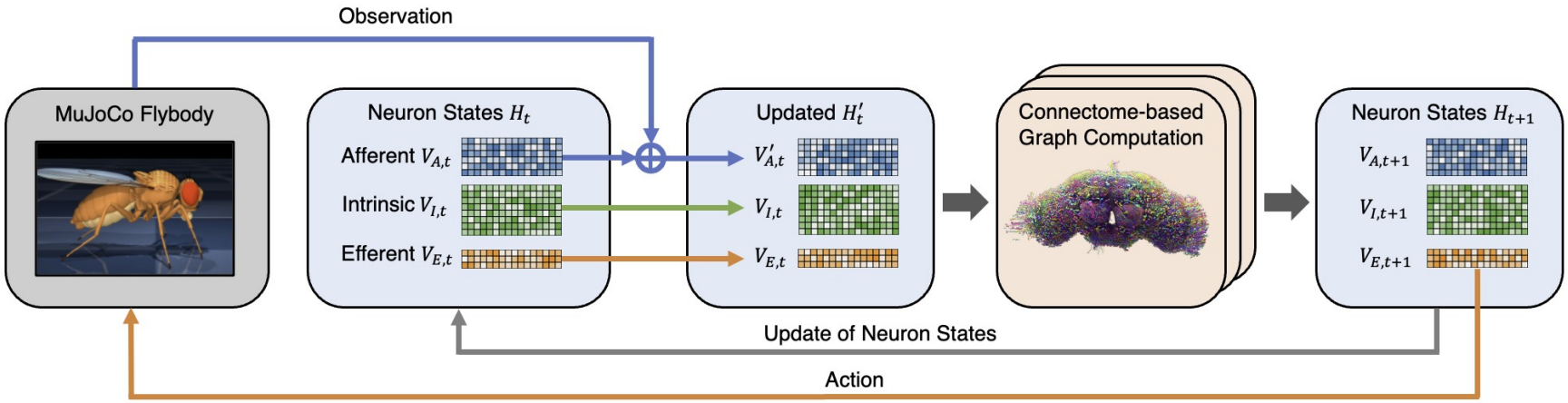
RESEARCH QUESTION

**Can whole-brain wiring serve as a useful structural prior for whole-body control?**

Images: FlyWire brain [2]; flybody fly model [7]; FlyGM force-directed connectome layout [1].

METHOD

# FlyGM closes the sensorimotor loop




## 1 Observe

Proprioception, touch and binocular vision (1,253-D) are encoded and injected into afferent neurons.

## 2 Propagate and update

Messages flow along the synapses of ~139k neurons, M = WH; a shared MLP updates each neuron with its own descriptor.

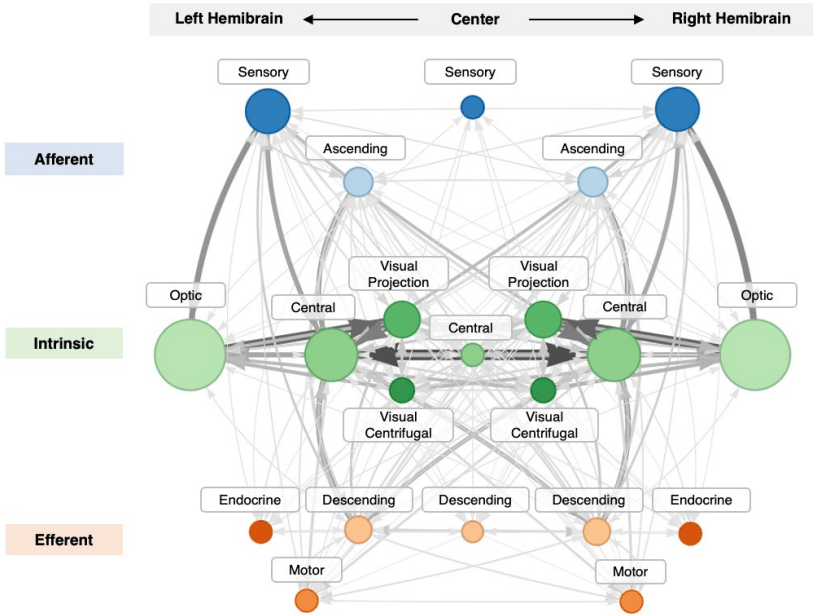
## 3 Act

Efferent states are decoded into 59 actuator commands for legs, adhesion, head and abdomen in MuJoCo.

FlyGM [1]: Fig. 1 and §2.1. Walking configuration shown; flight uses a 104-D input and 12 actions. Simulator: flybody [7] in MuJoCo [14].

METHOD

# The wiring is fixed; the computation is learned




*FlyWire neurons grouped by superclass and flow type; edges show aggregated synapse counts.*

## FIXED from the connectome

- **Directed topology:** who can talk to whom
- **Signed weights:** $W_{vu} = N_{exc} - N_{inh}$ (ACh, DA +; GABA, Glu, 5-HT, OA −) [12, 13]
- **Flow types:** afferent → intrinsic → efferent

## LEARNED by training

- Observation encoder and action decoder
- Per-neuron descriptor $\eta_v$ (e.g. excitability, gain)
- Shared state-update network $f_\psi$

$$M_t = W H_t \qquad H_{t+1}[v] = f_\psi( M_t[v] \,\|\, \eta_v )$$

FlyGM [1]: §2.1 and Appendix E. Latent states are learned vectors, not firing rates; signed weights are a modeling convention.

TRAINING

# Imitation gives a head start; PPO refines it

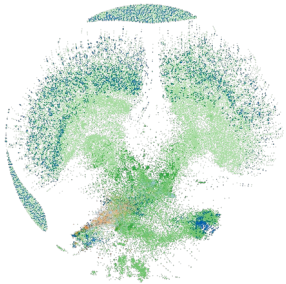
**Expert behavior**

flybody MLP policy [7]; rollouts kept if > 100 steps

1

**Imitation learning**

Match the expert's action distribution: KL + annealed MSE



**FlyGM policy**

Connectome-structured controller, initialized

2

**PPO refinement**

Optimize task reward in MuJoCo [15]; MLP value network, parallel rollouts

**Task interaction**

Walking, turning and flight tasks

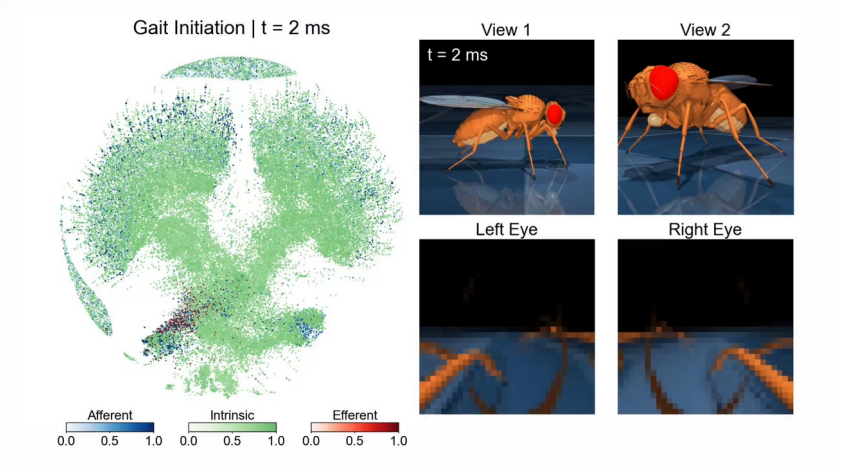
**Same IL + PPO pipeline for every controller we compare**

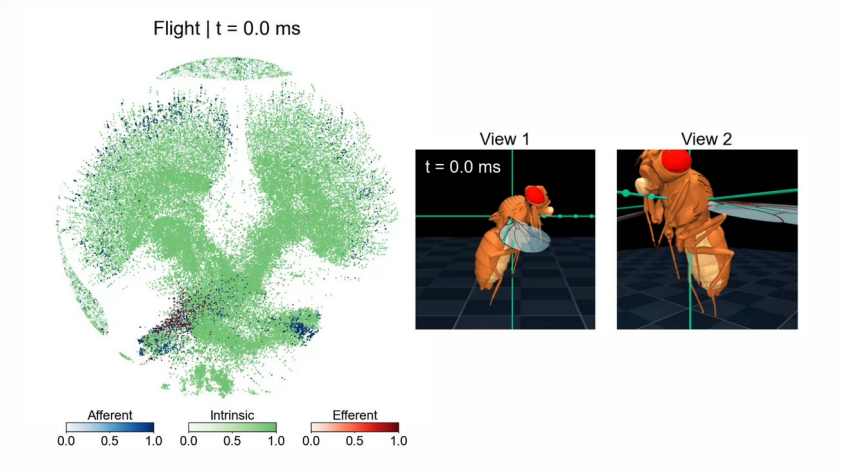
**Walking and flight use separately trained policies**

FlyGM [1]: §2.2 and Appendix B. Expert policies and environments from flybody [7].

BEHAVIOR

# FlyGM drives coordinated walking, turning and flight




## Walking policy

**Gait onset** ≈ 80 ms

**Tripod gait** 3 cm/s

**Turns** 10 rad/s

## Flight policy (trained separately)

**Forward flight** 20 cm/s

**Wing-beat generator modulation**

*Click a video to play. Playback is slowed for viewing; read the simulation timestamps.*

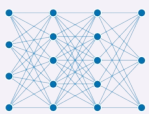
FlyGM [1]: §3.2, Figs. 3–4. Left panels: force-directed connectome layout with afferent, intrinsic and efferent activity.

# The real wiring learns faster and steers more accurately

**FlyGM**
Real FlyWire wiring with signed weights

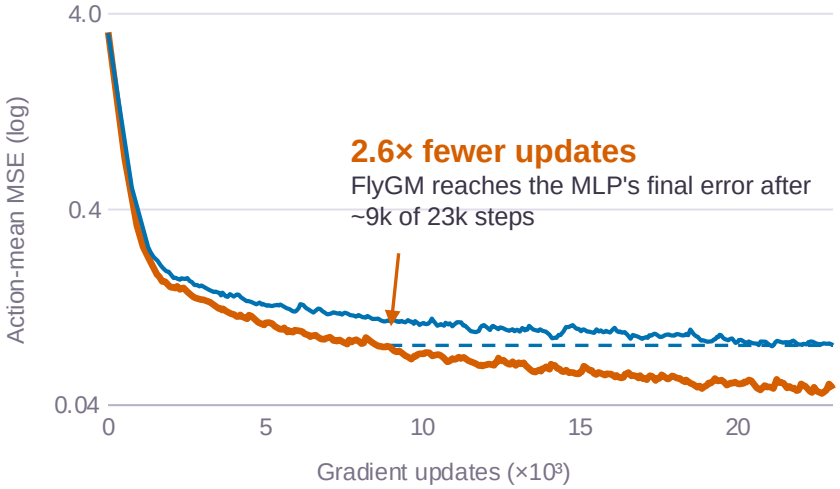
**DP-Rewired**
Same in/out-degree per neuron, shuffled partners

**ER-Random**
Same numbers of neurons and edges

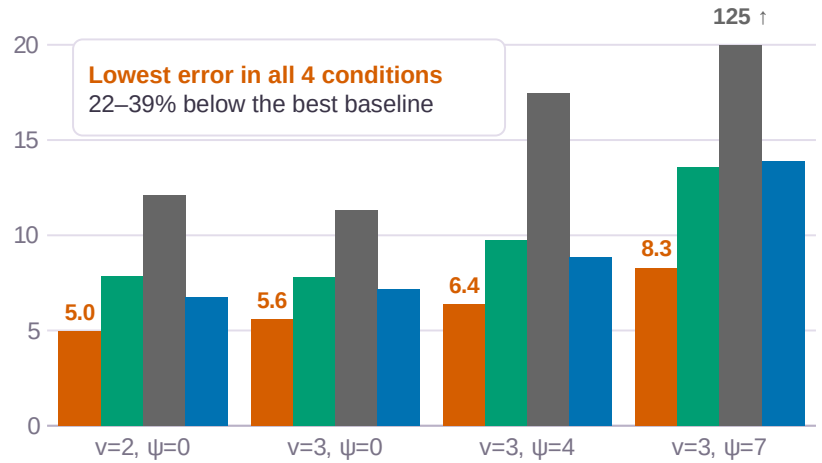


**MLP**
4 × 512 layers; more parameters than FlyGM

**Imitation: error to the expert's actions**




**After IL + PPO: heading-angle error (°)**



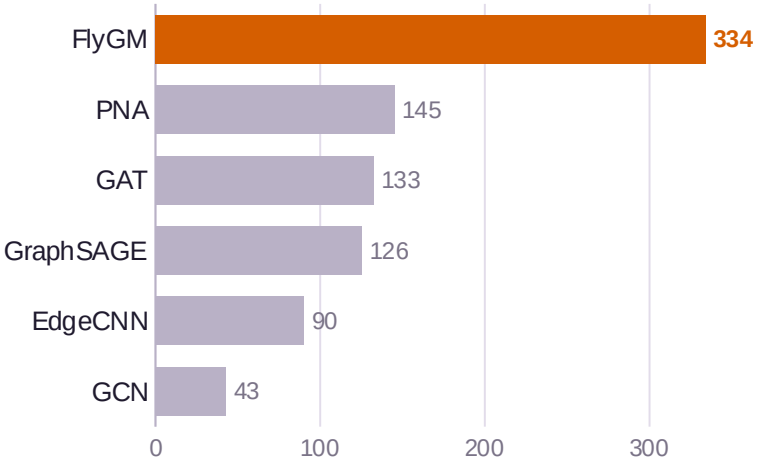

FlyGM [1]: Fig. 2a (one training trace per model, EMA) and Table 1 (10 evaluation rollouts of a fixed checkpoint; v in cm/s, ψ in rad/s).

RESULTS · WHAT DRIVES THE GAIN?

# The gain holds up under ablations and GNN baselines

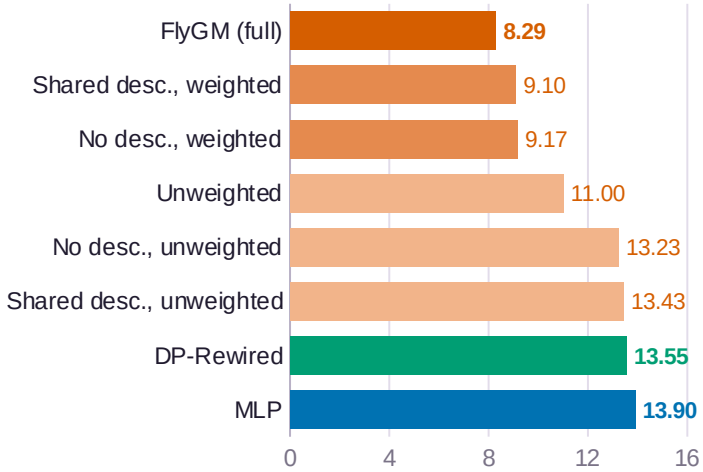
## Generic GNN layers fall short

Five standard GNNs on the same FlyWire neurons: average reward after IL + PPO




## Every connectome variant beats the controls

Ablations at the hardest condition (v = 3 cm/s, ψ = 7 rad/s): angle error (°)




**2.3×** the reward of the best standard GNN (PNA). A spiking network with the MLP layout never learns a stable gait.

**Topology carries most of the prior;** signed weights help most where control is hardest (8.3° vs 11.0° unweighted).

FlyGM [1]: Tables 2–4 (Appendix). GNNs [16–20] use the largest width and depth that fit one A100 80 GB; reward table repetitions not verified.

# One rollout links walking, pausing and turning

**Pause**

Side view

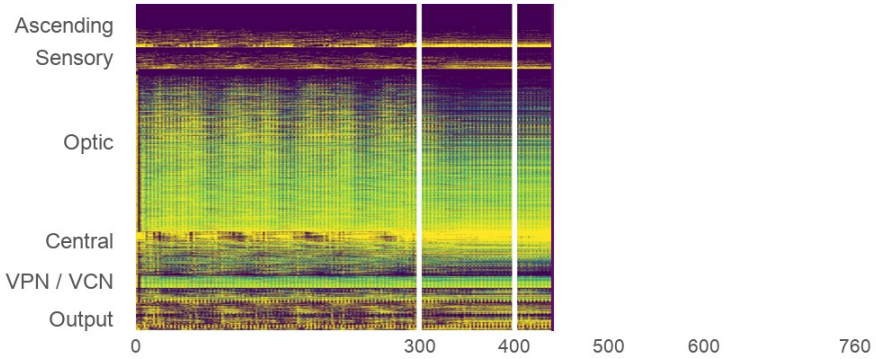
440 ms | 10× slower

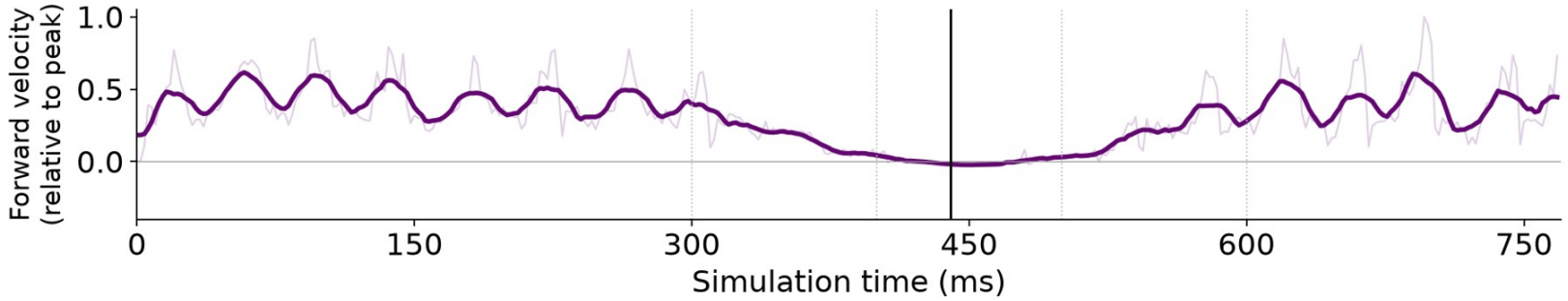
Front view

Same recorded sequence, two cameras

Model activity · paper Fig. 5

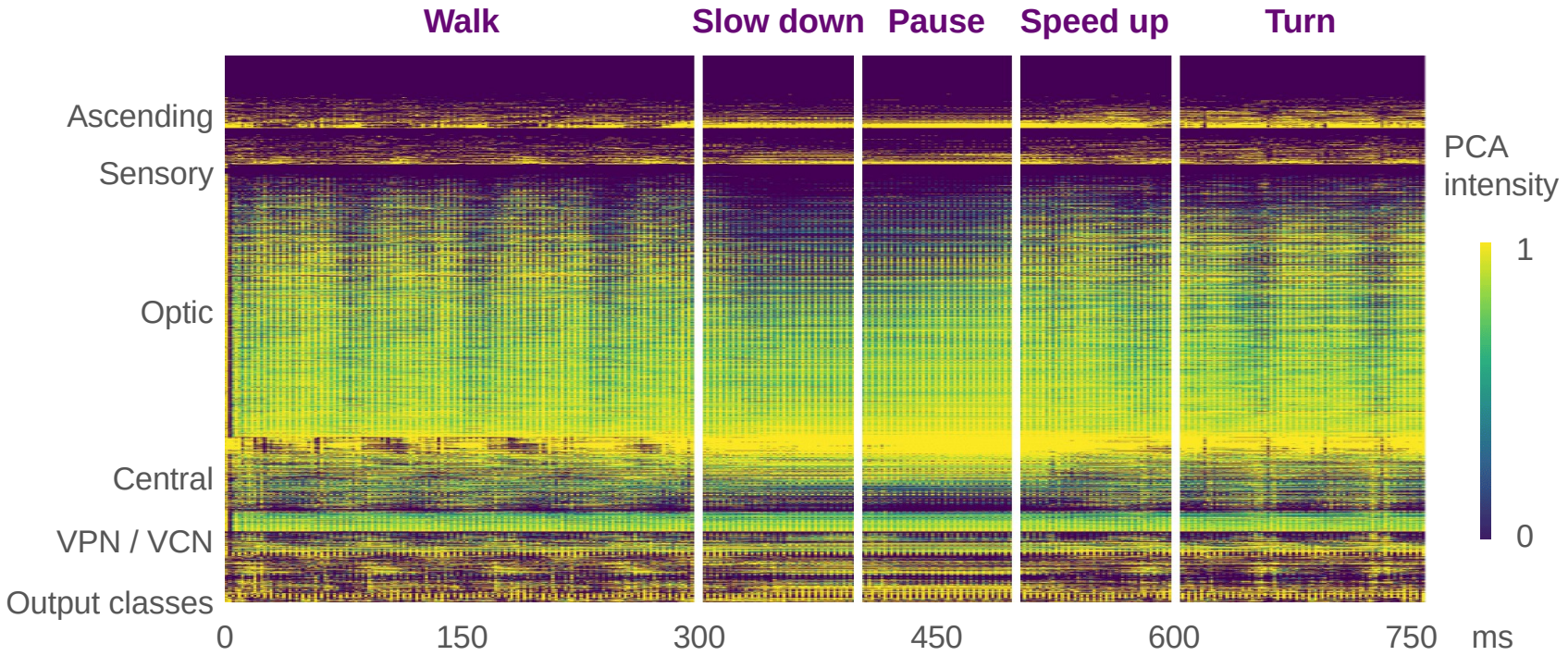
**440 ms**




Original rollout: 0–768 ms · click to play at 10× slow motion · illustrative archived recording; activity panel from FlyGM [1] Fig. 5.

# Latent responses differ across neuron classes

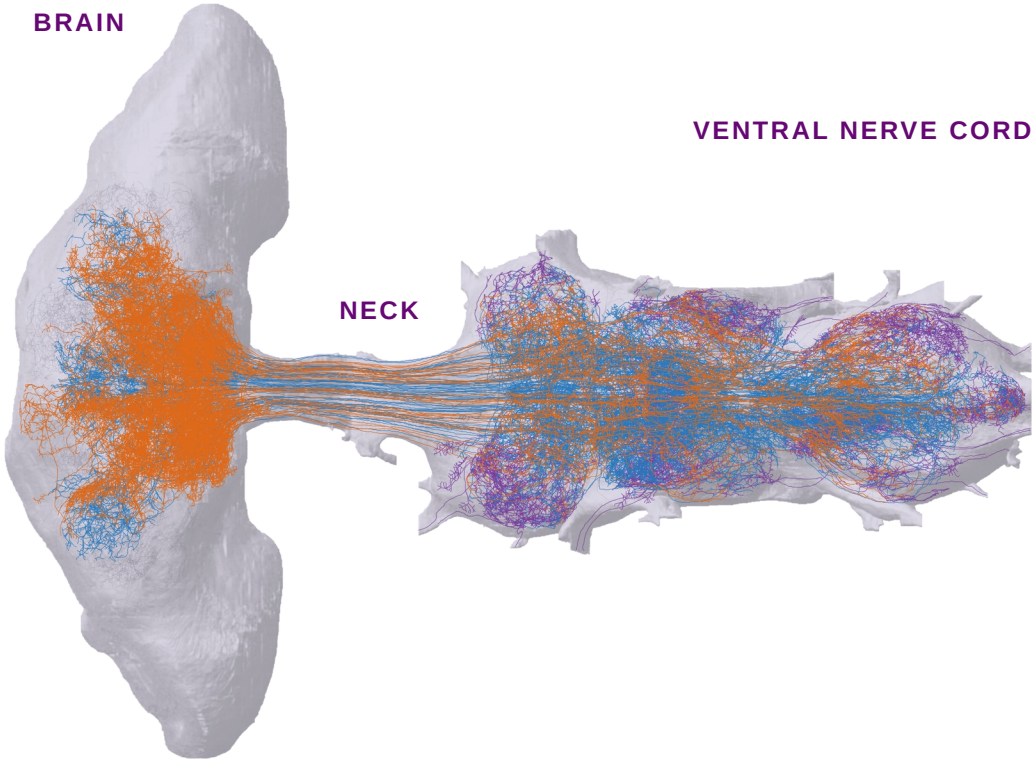



**Qualitative latent structure; physiological correspondence remains untested**

FlyGM [1]: Fig. 5 and Appendix F (PCA, class grouping, within-class ordering); random-graph control in Fig. 8. Cf. [21, 22].

OUTLOOK

# Next: from a brain-only graph to brain + nerve cord




Descending (brain → VNC) · Ascending (VNC → brain) · VNC motor neurons

**166,700**

neurons in the first complete male central nervous system: brain and ventral nerve cord [8]

**1,314** descending · **1,846** ascending neurons

### WHAT IT WOULD ADD TO FLYGM

- **Real premotor and motor circuits** between descending neurons and the legs
- **Ascending feedback** from legs and wings through measured pathways
- **Richer annotations:** updated synapses [23], sex-specific cell types [24]

**Next tests** longer sequences · perturbation recovery · navigation

Outlook, not a FlyGM result. Rendered from the public FlyEM male CNS v1.0 release: brain/VNC shells and ~280 sampled neurons [8].

TAKEAWAYS

# Brain wiring is a useful prior for whole-body control

1

## Wiring as architecture

The 139k-neuron FlyWire graph becomes a trainable whole-body controller for walking, turning and flight.

2

## The real wiring matters

Faster imitation and lower steering error than rewired, random and MLP controls; 2.3× the reward of standard GNNs.

3

## Structured dynamics

Neuron classes show distinct, behavior-locked latent patterns that a random graph lacks (qualitative).

**Not an upload, but a testable, connectome-structured controller.**

**Open:** biological fidelity of the learned dynamics · learning without demonstrations · other bodies and simulators

FlyGM [1]. Comparisons use the shared IL + PPO pipeline; representation analysis is qualitative (Appendix H).

# Thank you

Questions are welcome.

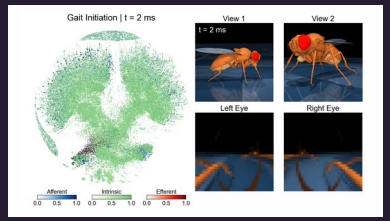
## SEE THE DEMOS

**lnsgroup.cc/research/FlyGM**

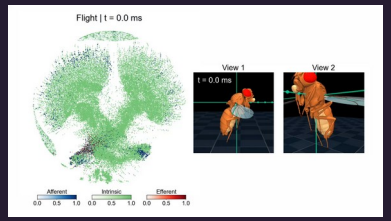
Walking, turning and flight videos with live connectome activity.

**Zehao Jin, Yaoye Zhu, Chen Zhang, Yanan Sui**
Tsinghua University · LNS Group

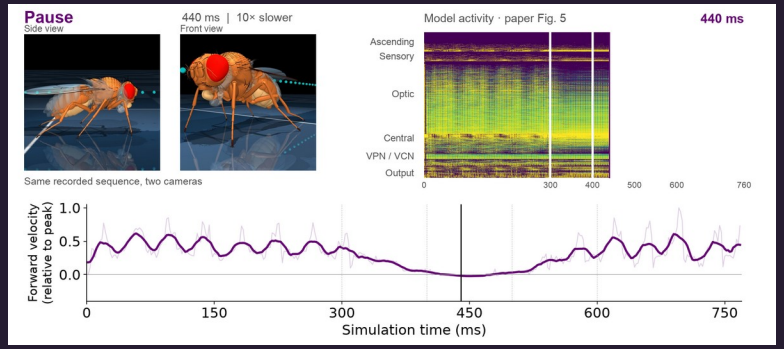
## STILLS FROM THE DEMO VIDEOS




Gait initiation and walking




Flight




Walk → pause → turn, with model activity

BACKUP

# References and resources

**DEMO PAGE**
lnsgroup.cc/research/FlyGM

**PAPER (PREPRINT)**
arXiv:2602.17997

**CONNECTOME DATA**
codex.flywire.ai · male-cns.janelia.org

**BODY SIMULATOR**
github.com/TuragaLab/flybody


**[1]** Z. Jin, Y. Zhu, C. Zhang, Y. Sui. Whole-brain connectomic graph model enables whole-body locomotion control in fruit fly. AIAS+ 2026; arXiv:2602.17997.

**[2]** S. Dorkenwald et al. Neuronal wiring diagram of an adult brain. Nature 634, 124–138 (2024).

**[3]** P. Schlegel et al. Whole-brain annotation and multi-connectome cell typing of Drosophila. Nature 634, 139–152 (2024).

**[4]** P. K. Shiu et al. A Drosophila computational brain model reveals sensorimotor processing. Nature 634, 210–219 (2024).

**[5]** J. K. Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature 634, 1132–1140 (2024).

**[6]** S. Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. Nat. Methods 21, 2353–2362 (2024).

**[7]** R. Vaxenburg et al. Whole-body physics simulation of fruit fly locomotion. Nature 643, 1312–1320 (2025).

**[8]** S. Berg et al. Sexual dimorphism in the complete connectome of the Drosophila male central nervous system. Cell 189, 5504–5526 (2026).

**[9]** Eon Systems. “We’ve uploaded a fruit fly”; “How the Eon team produced a virtual embodied fly”. eon.systems/updates (Mar 2026).

**[10]** A. Zeleznikow-Johnston. No, we haven’t uploaded a fly yet. LessWrong (Mar 2026).

**[11]** P. Mineault. Deconstructing viral fly sims. NeuroAI, neuroai.science (2026).

**[12]** D. A. Pospisil et al. The fly connectome reveals a path to the effectome. Nature 634, 201–209 (2024).

**[13]** N. Eckstein et al. Neurotransmitter classification from electron microscopy images at synaptic sites in Drosophila melanogaster. Cell 187, 2574–2594 (2024).

**[14]** E. Todorov, T. Erez, Y. Tassa. MuJoCo: a physics engine for model-based control. IROS, 5026–5033 (2012).

**[15]** J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347 (2017).

**[16]** T. N. Kipf, M. Welling. Semi-supervised classification with graph convolutional networks. ICLR (2017).

**[17]** Y. Wang et al. Dynamic graph CNN for learning on point clouds. ACM Trans. Graph. 38, 1–12 (2019).

**[18]** W. Hamilton, Z. Ying, J. Leskovec. Inductive representation learning on large graphs. NeurIPS 30 (2017).

**[19]** P. Veličković et al. Graph attention networks. ICLR (2018).

**[20]** G. Corso et al. Principal neighbourhood aggregation for graph nets. NeurIPS 33, 13260–13271 (2020).

**[21]** B. E. Brezovec et al. Mapping the neural dynamics of locomotion across the Drosophila brain. Curr. Biol. 34, 710–726 (2024).

**[22]** E. S. Schaffer et al. The spatial and temporal structure of neural activity across the fly brain. Nat. Commun. 14, 5572 (2023).

**[23]** S.-C. Yu et al. New synapse detection in the whole-brain connectome of Drosophila. bioRxiv (2025).

**[24]** D. Deutsch et al. Sexually-dimorphic neurons in the Drosophila whole-brain connectome. Research Square (2025).